

VitalsSight: A Candidate-Aware Framework and Auditable Output Contract for Contactless Heart Rate Monitoring

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Abstract

Camera-based contactless heart rate monitoring offers significant potential for clinical care and longitudinal research, yet estimation often fails in challenging windows containing competing pulse candidates, aliases, or motion artifacts. To overcome this deployment bottleneck, we present VitalsSight, a candidate-aware supervised framework that retains pulse candidates across diverse signal routes prior to selection, thereby preserving critical physiological evidence. In internal subject-disjoint evaluations on the University of Bourgogne Franche-Comté remote photoplethysmography (UBFC-rPPG) benchmark dataset (42 participants, 439 windows), VitalsSight achieved a primary window-level mean absolute error of 1.646 ± 0.051 beats per minute, with consistent error reductions across stress datasets including MCD-rPPG and rPPG-10. Crucially, when encountering real-world deployment boundaries—such as severe low light and source shifts—VitalsSight outputs an evidence packet capturing candidate ambiguity to support a transparent release/review contract. Rather than masking errors through forced automated outputs, this auditable mechanism establishes a safer and more pragmatic foundation for the clinical translation of digital vital sign monitoring systems. Code available at <https://github.com/ShikiRyo1/VitalsSight-candidate-aware-rPPG>.

Introduction

Heart rate is routinely monitored across clinical care, rehabilitation, and longitudinal research. Because traditional contact sensors can impede prolonged measurement, camera-based contactless monitoring has been extensively studied in intensive care, neonatal units, and less constrained settings^[1–4]. Remote photoplethysmography (rPPG) extracts pulse information from subtle facial-light changes, which are frequently obscured by motion, illumination, compression, and camera responses; furthermore, the specific choices of facial regions and color channels can substantially alter the captured physiological evidence^[5–8].

Clinical translation of rPPG requires both highly accurate vital sign estimates and a transparent withholding rule to handle periods when physiological support weakens^[9–11]. In practical deployments, a challenging video window may simultaneously contain the true pulse, a harmonic or alias, and a motion-favored noise candidate. While classical blind-source separation, chrominance-based methods, and spatial-redundancy routes encode different nuisance assumptions^[12–15], motivating the simultaneous retention of multiple plausible signal routes—modern deep learning pipelines such as DeepPhys, TSCAN, and EfficientPhys introduce learned spatial-temporal representations. However, these network architectures remain highly sensitive to training dynamics, preprocessing protocols, and video acquisition domains^[16–18]. Although PhysFormer and RhythmFormer have successfully extended this domain using temporal difference and periodic transformers^[19–20], and open toolboxes have significantly improved framework transparency^[21–22], no single route is uniformly reliable. This inherent vulnerability strongly motivates separating candidate generation from subsequent route selection and withholding.

Prior tracking techniques focus on constraining a heart rate estimate after it has already been formed, yet post-hoc track correction fundamentally cannot restore a valid physiological branch once it has been prematurely discarded by an upstream backbone^[23]. Routing heterogeneous predictors to solve this issue closely resembles mixtures of local experts and stacked generalization^[24–25]; therefore, establishing candidate-aware selection requires rigorous comparisons against simpler gates, consensus mechanisms, and stacking methods.

Despite these extensive developments in both classical and deep SOTA frameworks, a critical research gap persists: existing pipelines typically force a single signal route execution or rely on post-hoc path corrections, lacking an auditable withholding mechanism when physiological evidence degrades. This lack of transparency under extreme noise, source shifts, or low light severely limits the safe integration of rPPG into high-stakes clinical workflows.

To bridge this gap and separate prediction from autonomous release^[26–29], we introduce VitalsSight, a highly practical supervised framework that reframes rPPG estimation failure as candidate ambiguity. By decoupling candidate generation from relation-aware selection, our study delivers the following core scientific contributions:

1. **A Candidate-Aware Inference Architecture:** Unlike forced single-route predictors, VitalsSight dynamically models candidate ambiguity as supervised selection with explicit review, retaining competing heart rate candidates from multiple classical, deep, and transformer families prior to final selection to preserve critical physiological evidence.
2. **An Auditable Release/Review Contract:** The framework ranks candidate-relation evidence using only inference-time features—strictly excluding reference labels and error metrics from runtime inputs—and returns

a detailed evidence packet. This architecture directly supports the transparent withholding rules required for safe clinical translation by explicitly surfacing model ambiguity rather than masking errors.

3. **Rigorous Boundary and Stress Auditing:** Through subject-disjoint internal evaluation, source decomposition, and cross-dataset stress testing (including severe low-light conditions and synthetic failure domains), we systematically map the boundaries and transportability limits of the proposed selector, establishing a safer and more pragmatic foundation for digital medical monitoring systems.

Results

Study design and evidence hierarchy

To address the clinical bottleneck of silent algorithmic failures in remote photoplethysmography (rPPG), we developed VitalsSight, an auditable framework structured around a candidate-aware inference pipeline and an explicit output contract (Figure 1). As conceptualized across three interconnected panels, Figure 1a delineates the candidate construction and relation-aware selection phase, where observable video/ROI, temporal, and spectral evidence enter five distinct route families. This populates a variable-size candidate set, where each record explicitly preserves its estimated heart rate, quality, harmonic support, route identity, and validity. Cross-candidate relational topologies expose underlying consensus or divergence via relation edges before selection. The labelled interface carries the selected estimate and full candidate context unchanged into Figure 1b, where the system independently computes an empirical evidence packet and a research score before the proposed output contract maps the window into either a "release" or a "review" state. Crucially, Figure 1c establishes a strict operational wall that separates inference-time evidence from evaluation-only references and unresolved deployment claims, guaranteeing the system's runtime audibility. Panels a and b are therefore consecutive parts of one mechanism, ensuring that the selected value, its supporting evidence packet, and the proposed release state remain structurally distinct.

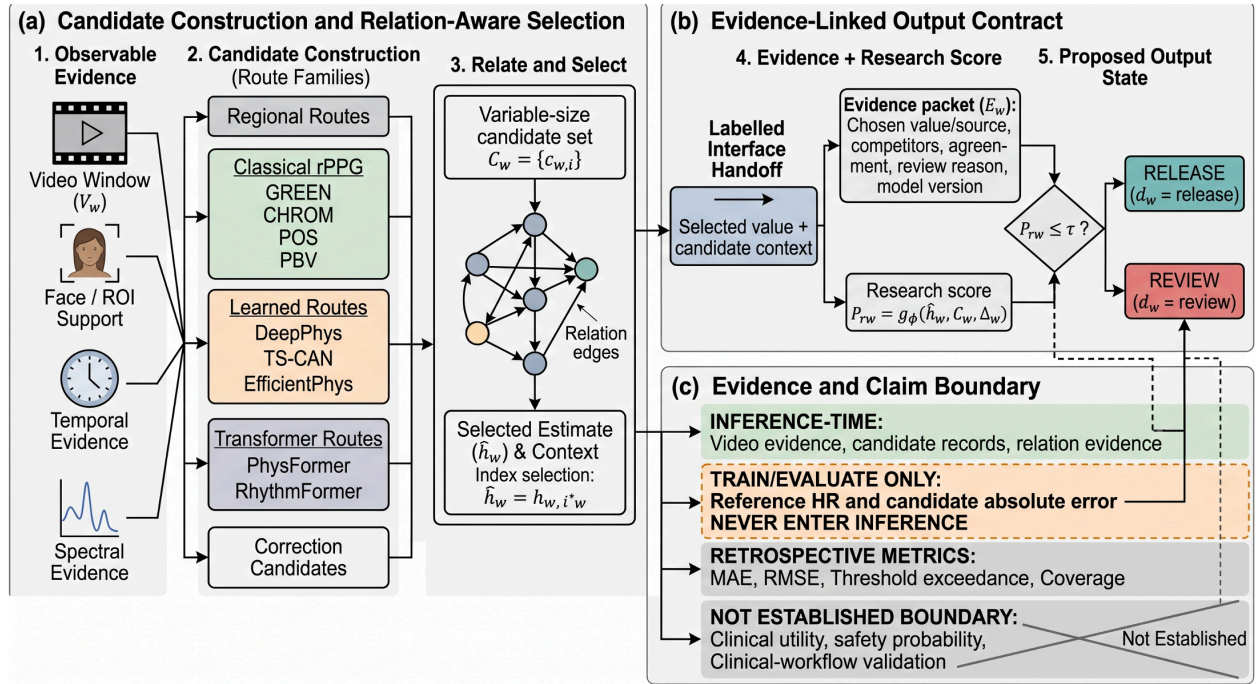


Figure 1 | Candidate-aware VitalsSight inference and proposed review contract. (a) Observable video/ROI, temporal and spectral evidence enters five route-specific candidate families; a variable-size candidate set, and relation-aware score select one estimate while retaining source identity. (b) The labelled handoff carries the selected estimate and candidate context into separate evidence-packet and research-score branches, which jointly propose release or review without merging value, evidence and state into one safety claim. (c) Reference heart rate and candidate error remain training/evaluation-only; the research score is not a calibrated safety probability or clinical-workflow validation.

To rigorously validate the VitalsSight framework and test its source complementarity and withholding mechanisms, we systematically assigned datasets and protocol branches into primary within-dataset internal performance evaluation, mechanism or boundary roles. As detailed in Table 1, the University of Bourgogne Franche-Comté remote photoplethysmography (UBFC-rPPG) dataset served as the primary internal benchmark, encompassing 42 participants and 439 analyzed windows evaluated under a strict subject-disjoint protocol. Physiological and environmental stress robustness were audited using MCD-rPPG (post-exercise and high-HR stress), a classical-pool variant of UBFC-rPPG, and rPPG-10. Transportability and failure boundaries under source shifts, synthetic anomalies, severe low light, and subgroup accessibility constraints were established using UBFC-Phys, SCAMPS-full, rPPG-OSF, MR-NIRP, CMU proxies, and DLCN. Detailed mapping of sampling hierarchies, random seeds, subject-equal summaries, and threshold crosswalks are preserved in Supplementary Tables S1-S2.

Table 1 | Datasets, sampling hierarchy and evidence roles

Dataset	Independent units	Windows/records	Role in this study	Interpretive boundary
UBFC-rPPG	42 subjects	439 windows	Primary within-dataset subject-disjoint internal evaluation	Not an independent external validation; windows are repeated within subjects
MCD-rPPG	6 subjects	60 windows	Post-exercise and high-HR stress analysis	Official paper and CC BY 4.0 dataset record verified; publication consent for identifiable faces is not publicly documented; release-protocol variants retained separately
UBFC-Phys S1-S14	5 subjects	45 windows	Source-shift boundary	Conflicting release protocols preclude a positive claim
rPPG-10	26 subjects (canonical tier)	26 subject-level units	External ROI-level descriptive analysis	Participant-record map pending canonical-source migration
4TU-rPPG Benchmark	21 sessions	21 sessions	External-video bridge stress analysis	Not a standalone SOTA claim
SCAMPS-500 / full	Synthetic	500 / 2,800 samples	Synthetic sanity and failure boundary	Does not represent human participants or clinical transportability
rPPG-OSF	42 signal-level records	42 records	Camera/source-shift boundary	Descriptive reduction; large error tail; no inferential claim
MR-NIRP	8 participants/15 released sequences (separate case audit)	Retained summary denominator separately unresolved	Low-light boundary	Case audit does not replace retained summary; not a demographic label
CMU proxy	88 group-labelled records	88 records	Subgroup-assessability audit	Site/country proxy is not skin tone
DLCN	Dynamic-lighting candidate pool	Scale pending source migration	Candidate-pool mechanism boundary	Oracle gap is not deployable performance

Subject counts are reported separately from windows or records. Boundary denotes failure or limited transportability, not positive external validation.

Primary internal performance and candidate source decomposition

The primary internal evaluation of the set-attention selector, incorporating temporal correction candidates, was conducted on 439 signal windows derived from 42 participants within the UBFC-rPPG dataset. Across three independent model initialization seeds (704, 1704, and 2704), the framework achieved a highly stable window-level mean absolute error (MAE) of 1.646 ± 0.051 beats per minute (BPM) and a root mean square error (RMSE) of 4.946 ± 0.186 BPM. The system successfully constrained $96.8 \pm 0.0\%$ of all estimations within an analytical 10 BPM error margin, with only $3.2 \pm 0.0\%$ of windows exceeding this threshold (Table 2). The retained subject-equal MAE was 1.662 BPM. It is critical to note that the reported dispersion reflects algorithmic variance across model seeds rather than participant-level confidence intervals, as matched learned-baseline reruns and paired cluster analyses remain incomplete.

Table 2 | Candidate-source decomposition in the primary UBFC-rPPG internal protocol

Source route	Single-route MAE	Error >10 BPM (%)	Selected frequency	MAE when selected
Classical max-SNR	45.738	92.9%	0.0%	N/A
PhysFormer	17.906	46.2%	6.5%	3.722
EfficientPhys	12.189	27.3%	3.8%	0.844
RhythmFormer	8.937	18.2%	13.9%	4.034
DeepPhys	6.535	15.5%	65.5%	0.171
TSCAN	5.990	13.9%	3.0%	1.623
Other classical/correction routes	N/A	N/A	7.4%	Not reported
VitalsSight full candidate selector	1.646	3.2%	N/A	N/A

Frequencies use 1,317 seed-window decisions ($439 \text{ windows} \times 3 \text{ seeds}$), not independent observations. Other combines six non-equivalent low-frequency routes: $98/1,317 = 7.4\%$; rounded rows may total 100.1%. It has no route-specific conditional MAE and is Not reported. The full selector is a method aggregate, so frequency and conditional MAE are N/A. These are unmatched source audits; best-value bolding is not used.

To elucidate the framework's underlying candidate-selection hypothesis, Figure 2 provides a synthetic mechanism schematic of candidate ambiguity. This visualization demonstrates how competing simulated spectral branches (Figure 2a) are evaluated using cross-route agreement parameters (Figure 2b) to inform the final selected trace (Figure 2c). Crucially, all values depicted in Figure 2 are strictly illustrative and are excluded from quantitative evaluations; the comparator trace is provided solely to clarify retrospective evaluation mechanics and does not function as an inference-time input. Because classical spatial and chrominance routes react heterogeneously to mixture structures, retaining a diverse candidate pool is theoretically motivated, though this synthetic schematic alone does not prove empirical selector superiority.

To empirically determine whether the system's aggregate performance was driven by the dominance of a single deep backbone, we conducted a comprehensive candidate-source decomposition. As detailed in Table 2 and Figure 3a, standalone local route executions yielded substantially higher window-level MAEs: 45.738 BPM for the classical max-SNR route, 17.906 BPM for PhysFormer, 12.189 BPM for EfficientPhys, 8.937 BPM for RhythmFormer, 6.535 BPM for DeepPhys, and 5.990 BPM for TSCAN.

Rather than defaulting to the single best-performing baseline, the relation-aware selector dynamically leveraged the heterogeneous pool across 1,317 seed-window decisions (Figure 3b). DeepPhys was selected most frequently at 65.5% (yielding a conditional MAE of 0.171 BPM when selected), followed by RhythmFormer at 13.9% (conditional MAE 4.034 BPM), PhysFormer at 6.5% (conditional MAE 3.722 BPM), EfficientPhys at 3.8% (conditional MAE 0.844 BPM), and TSCAN at 3.0% (conditional MAE 1.623 BPM) (Table 2). An aggregate of six other non-equivalent low-frequency classical and correction routes contributed the remaining 7.4% of decisions.

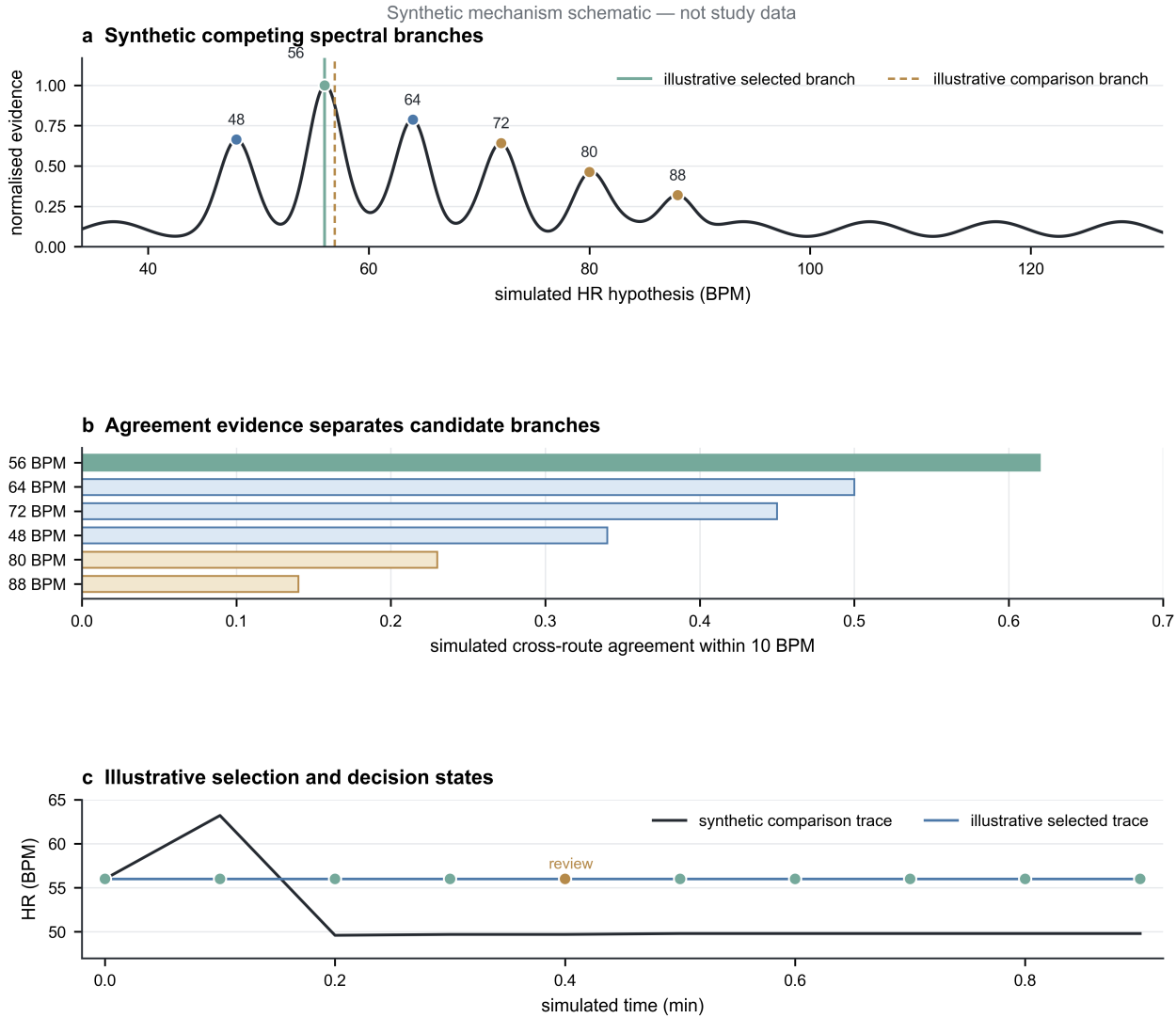


Figure 2 | Synthetic mechanism schematic of candidate ambiguity and auditable selection. Simulated spectral branches, cross-route agreement and an illustrative selected series explain the candidate-selection hypothesis; every plotted value is synthetic and excluded from quantitative analyses. The comparator trace is shown only to clarify retrospective evaluation and is not an inference-time gate input.

This decomposition explicitly separates protocol-bound source error from conditional route utilization. However, because the selector accesses multiple route outputs with asymmetric training histories, these results constitute unmatched source audits rather than strictly paired literature benchmarks. Consequently, the performance contrast between the full selector (1.646 BPM) and the top-performing local single route (TSCAN at 5.990 BPM) demonstrates structural complementarity within the candidate pool, rather than definitive algorithmic superiority over matched baselines. Extensive local route parameters and correction/gating audits supporting this structural complementarity are preserved in [Supplementary Tables S10a](#) and [S10b](#).

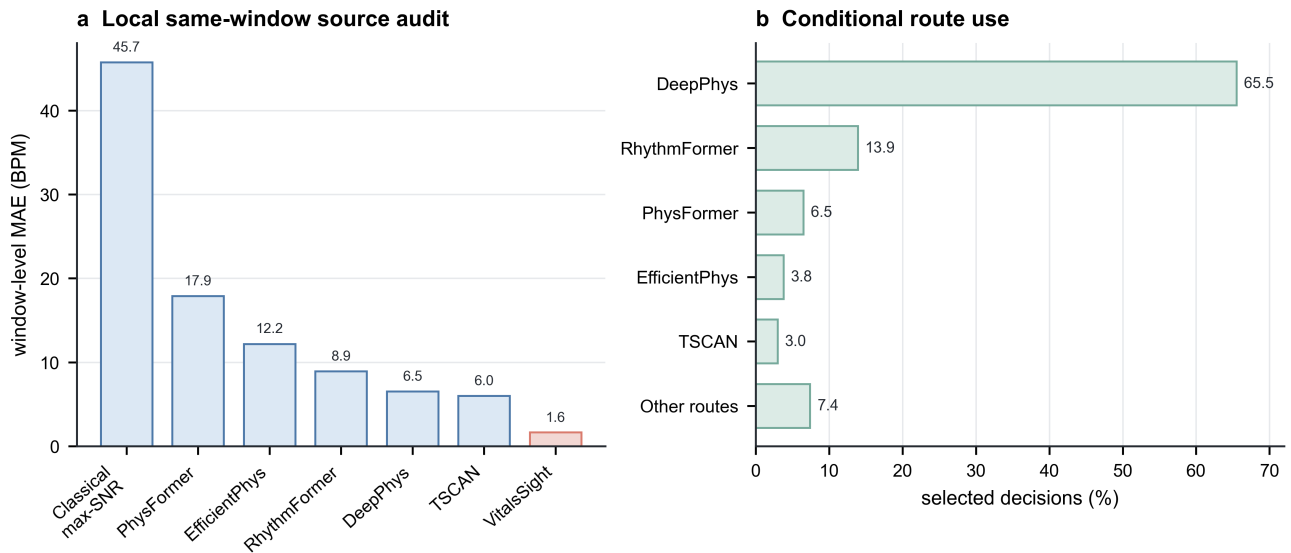


Figure 3 | Primary UBFC-rPPG candidate-source audit. (a) Window-level MAE for local same-window candidate routes and the full selector. The rows are protocol-bound source audits rather than matched literature benchmarks. (b) Conditional route-selection frequency across recorded seed-window decisions. Frequencies show that multiple routes contributed to the pool; they do not isolate the selector from ordinary ensemble or gating effects.

Exploratory abstention and risk analysis

A critical requirement for the clinical translation of remote vital sign monitoring is the capacity to transparently withhold predictions when physiological evidence degrades. However, our release analysis was designed to evaluate the framework's output contract as an auditable schema, not to establish a calibrated clinical safety gate.

We evaluated the abstention behavior of the VitalsSight output contract in a branch distinct from the primary 1.646 BPM protocol. [Figure 4](#) illustrates the fixed-coverage release characteristics across the recorded UBFC-rPPG cohort. Paradoxically, as the target coverage was artificially restricted to simulate higher-confidence thresholds, overall performance deteriorated. [Figure 4a](#) demonstrates that the released subset means absolute error (MAE) escalated from 1.654 ± 0.069 BPM at full coverage to 3.602 ± 0.481 BPM when retaining only the top 30% of windows. Concurrently, [Figure 4b](#) shows that errors exceeding the 10 BPM threshold doubled from $3.5 \pm 0.3\%$ to $7.1 \pm 1.2\%$. Because both endpoints worsened as coverage decreased, this curve serves as negative diagnostic evidence: the confidence scoring

does not preferentially filter out high-error windows, precluding any claims of beneficial abstention or automated risk control.

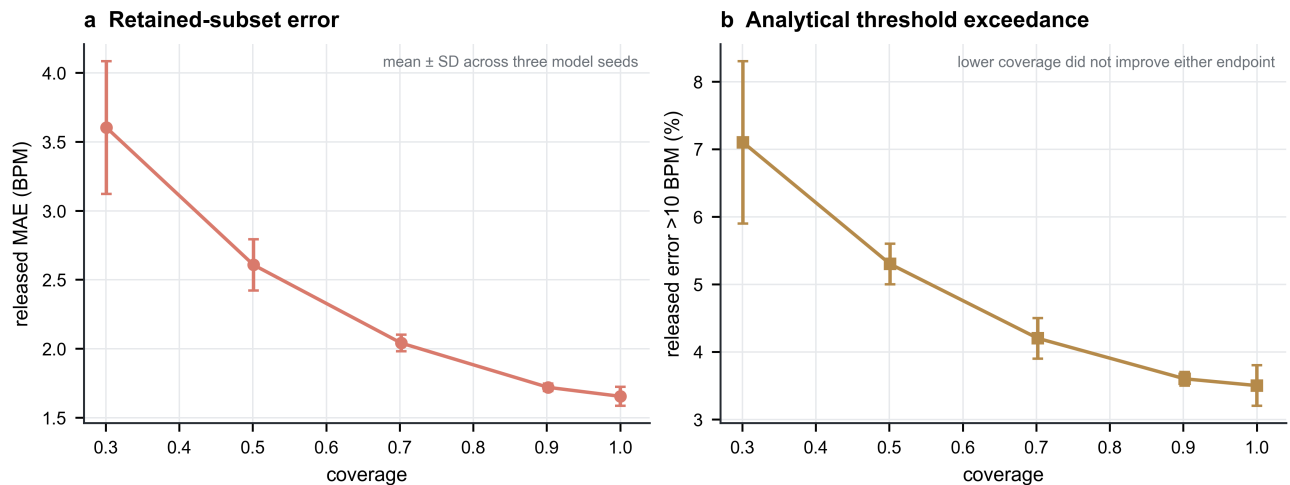


Figure 4 | Fixed-coverage release behavior in the recorded UBFC-rPPG branch. (a) Released-subset MAE and (b) released errors above 10 BPM are mean $\pm$ SD across seeds 704, 1704 and 2704. Per-seed points are absent from the frozen source package, while aggregates remain unchanged. Both endpoints worsened as coverage decreased; the curve is negative diagnostic evidence, not validation of beneficial abstention or risk control.

To interrogate this further, an entropy-margin rule was audited across three model seeds and three deterministic 14/14/14-participant splits derived from the same 42-person cohort. The mean and minimum held-out coverages were 98.8% and 94.6%, respectively, yielding a mean released error above 10 BPM of 3.35%. At an alpha level of 0.09 and a delta of 0.05, a naive one-sided binomial upper bound was calculated as 0.084. As detailed in [Supplementary Tables S7–S9](#), this calculation treats highly correlated video windows as independent observations; consequently, it functions strictly as an exploratory, post-hoc window-level diagnostic rather than a reliable participant-level guarantee.

Rigorous risk-control theory emphasizes the strict separation of model fitting, calibration, and release decisions. Because the risk-model form, coverage floors, and alpha thresholds were inspected retrospectively in our analysis, we applied Bonferroni adjustments to account for multiple hypothesis testing. The adjusted maxima rose to 0.107 across an eight-alpha sweep and 0.128 across a 64-point alpha-by-coverage grid. Neither configuration supports a strict familywise 0.09 risk claim. Furthermore, as documented in [Supplementary Note 3](#), a separate historical calculation ($\alpha = 0.10$, $\delta = 0.05$) required a minimum of 29 released windows per subset to achieve significance, even if zero errors above 10 BPM were observed. While established risk-control frameworks validate theoretical bounds, they do not intrinsically validate the window-level statistical unit utilized here, keeping this calculation explicitly separate from the primary analysis.

Finally, threshold sensitivity analyses exposed an additional protocol boundary. The proportion of errors exceeding 5 BPM was 16.1% under the primary protocol, compared to 19.6% under a separate split-calibration estimand. These heterogeneous values are preserved separately without averaging. It is imperative to note that the thresholds evaluated in this study (5, 8, 10, and 15 BPM) operate solely as analytical settings for framework benchmarking, not as actionable clinical limits. Establishing a formally ordered clinical risk requires a prospectively locked, participant-level test.

Recognizing this limitation in the current risk model, our subsequent investigations focused on architectural component sensitivity and cross-dataset stress boundaries.

Agreement and ablation sensitivity

We retained a comprehensive agreement analysis within the main manuscript, recognizing that aggregate MAE alone is insufficient to illustrate how selection, temporal correction, and gating alter systematic bias and error dispersion. Consequently, Figure 5 stratifies the seed-704 graph-selector branch into three distinct analytical categories to evaluate the impact of these components: all validation windows, initially high-error windows (those falling outside the analytical 10 BPM band prior to correction), and the final released subset.

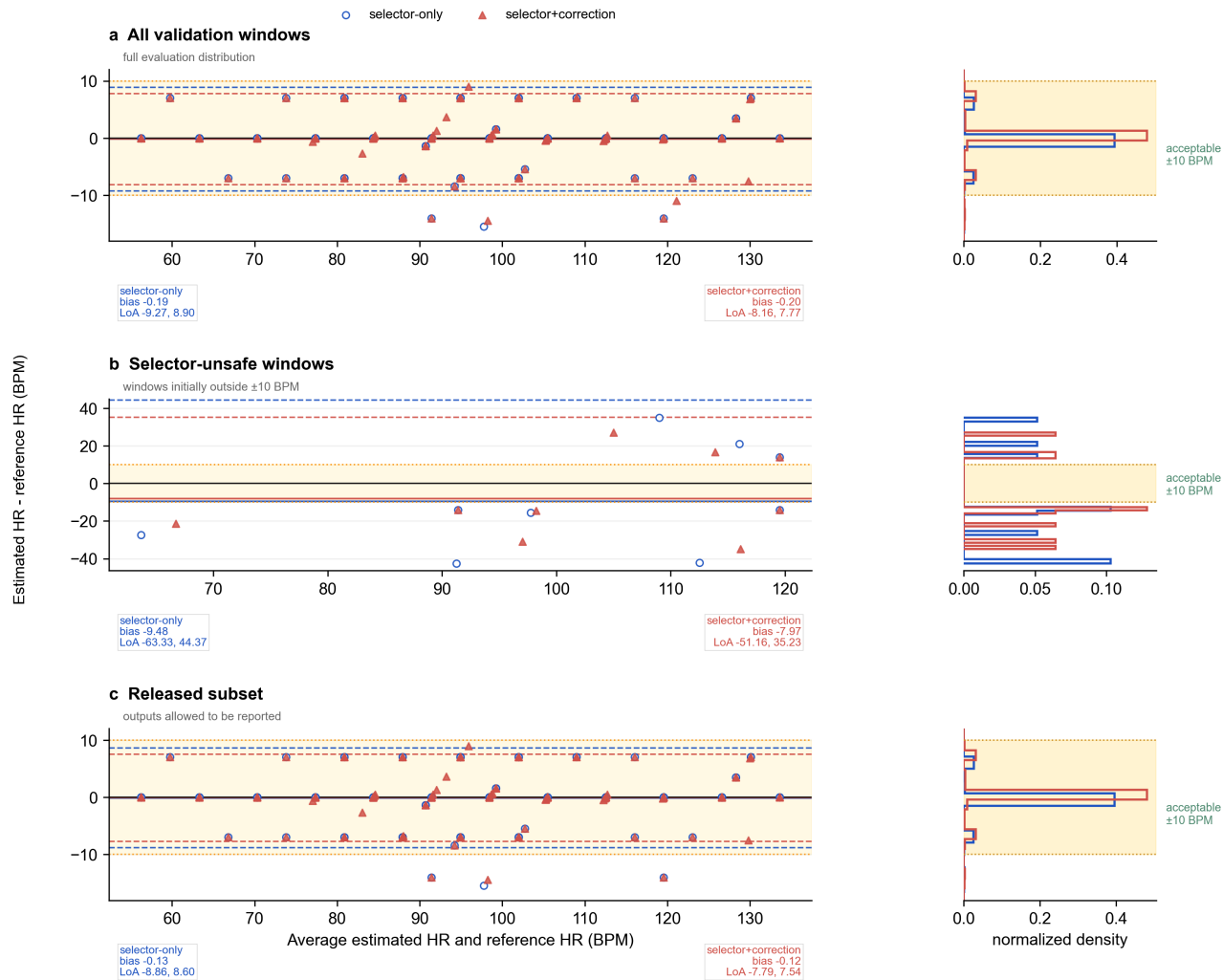


Figure 5 | Descriptive agreement analysis for the seed-704 graph-selector branch. (a) All windows, (b) windows outside 10 BPM before correction and (c) the released subset compare selector-only with selector-plus-correction estimates. Correction modestly narrows the displayed limits but leaves wide tails. Points are repeated windows, and the 10 BPM band is analytical rather than clinical. Python reprocessed the retained author composite without changing or inferring displayed values; point-level regeneration remains pending.

Evaluating all validation windows (Figure 5a), the selector-only baseline yielded a mean bias of -0.19 BPM with limits of agreement (LoA) spanning from -9.27 to 8.90 BPM. The integration of cross-fit correction candidates

marginally shifted the bias to -0.20 BPM while effectively narrowing the LoA to [-8.16, 7.77] BPM. For the initially high-error subset (Figure 5b), the correction mechanism demonstrated notable utility by reducing the systemic bias from -9.48 BPM down to -7.97 BPM. Furthermore, it substantially compressed the extreme error tails, improving the initial LoA from [-63.33, 44.37] BPM to [-51.16, 35.23] BPM. Within the final released subset (Figure 5c), the combined pipeline achieved highly stable biases of -0.13 BPM (selector-only) and -0.12 BPM (selector-plus-correction). The correction layer further tightened the LoA from [-8.86, 8.60] BPM to [-7.79, 7.54] BPM. These trends indicate that while temporal correction modestly narrows descriptive limits and successfully mitigates certain deviations post-selection, it cannot fundamentally repair an inherently incorrect physiological branch.

It is important to emphasize that these repeated-window limits serve as a descriptive diagnostic rather than formal participant-level agreement estimates. Across model seeds 704, 1704, and 2704, this specific analytical branch yielded an all-window MAE of 1.459 ± 0.090 BPM and a released MAE of 1.418 ± 0.115 BPM. However, these metrics represent an exploratory branch and do not replace the primary 1.646 BPM point estimate reported earlier.

To further contextualize mechanism behavior under ambiguity, Supplementary Figures S3 and S4 pair provider-released MR-NIRP frames with their matched candidate selections, synchronized PulseOx references, and final review decisions. Although near-reference candidates existed within the pool, all displayed sequences lacked a clean RGB/NIR cross-modal consensus and were therefore conservatively held in the review state. These descriptive case studies vividly illustrate the withholding mechanism's behavior but do not autonomously establish participant-level risk control.

We subsequently conducted structural ablations to determine the necessity of individual selector components, though the results did not establish strict component necessity. As detailed in Supplementary Figure S1, substituting identity graph edges degraded performance, increasing the MAE to 1.572 BPM. Removing either the harmonic penalty or the KL soft target resulted in an MAE of approximately 1.49 BPM, compared to 1.454 BPM for the complete graph selector architecture. Conversely, an alternative branch yielded an MAE of 1.499 ± 0.107 BPM without correction candidates but worsened to 1.654 ± 0.069 BPM when they were included. MCD-rPPG specific ablations remain preserved under a distinct protocol key pending final source-map reconciliation. Ultimately, these mixed ablation outcomes, coupled with the adverse risk-coverage diagnostic trend discussed previously, necessitated the rigorous stress and source-shift analyses presented in the following section.

Stress and generalization boundaries

Conventional fixed spectral or region-of-interest (ROI) baselines inherently commit to a single computational route, leaving them highly vulnerable to signal degradation when subject motion, elevated heart rates, or environmental source shifts destabilize the dominant pulse peak. VitalsSight circumvents this limitation by preserving competing candidate alternatives. To quantify this resilience, we subjected the framework to rigorous cross-dataset stress protocols, capturing performance retention beyond the primary internal baseline.

As detailed in Table 3, VitalsSight demonstrated substantial error reductions across multiple physiological and algorithmic stress conditions:

- **Elevated Heart Rate and Exercise Stress (MCD-rPPG):** Under challenging post-exercise conditions, the framework reduced the subject-equal mean absolute error (MAE) from a baseline of 24.242 BPM to 17.231 BPM. Utilizing the output contract, the released-subset MAE dropped to 7.928 BPM at a 45.0% coverage rate, retaining an error exceedance of only 7.4%.
- **Classical Candidate-Pool Stress (UBFC-rPPG classical):** When restricted to a purely classical candidate pool, VitalsSight reduced the window-level MAE from 45.738 BPM to 8.774 BPM (subject-equal MAE: 8.746 BPM). At 74.9% coverage, the released MAE stabilized at 5.971 BPM.
- **Canonical Subject-Level Auditing (rPPG-10):** The framework reduced the aggregate MAE from 10.164 BPM to 7.136 BPM, achieving a highly accurate released MAE of 2.102 BPM at 76.9% coverage, with a 0.0% released error exceedance above 10 BPM.
- **External Video Bridge (4TU-rPPG):** The system maintained a stable 4.009 BPM MAE and 4.875 BPM RMSE under full coverage conventions, with zero exceedance.

Despite these improvements, heterogeneous dataset protocols and incomplete coverage distributions preclude broad, generalized claims of external transportability. For instance, a localized audit of the MCD-rPPG dataset (retaining 96,983 rows) identified the IriunWebcam as the lowest-performing camera (median SNR -2.025) and confirmed the post-exercise phase as the highest-rate boundary. However, the retained 0.085 high-rate-gap scalar remains uninterpreted pending complete source migration. [Table 3](#) preserves all locked metrics, explicitly separating statistical units (e.g., subject-equal endpoints for MCD-rPPG and UBFC-Phys vs. synthetic samples for SCAMPS) to prevent inappropriate pooling of heterogeneous protocols.

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Table 3 | Protocol-specific metrics in cross-dataset stress and boundary analyses

Dataset/protocol	Method	Unit	MAE	RMSE	Within 10 BPM	Coverage	Released MAE	Error >10 BPM	Evidence role
MCD-rPPG stress	Max-SNR baseline	6 subjects (60 windows); subject-equal endpoint	24.242	33.673	38.3%	N/A	N/A	N/A	Post-exercise/high-HR comparator
UBFC-Phys S1-S14	Max-SNR baseline	45 windows / 5 subjects	19.665	25.654	42.2%	N/A	N/A	N/A	Source-shift comparator
UBFC-rPPG classical stress	Max-SNR baseline	439 windows / 42 subjects	45.738	52.621	7.1%	N/A	N/A	N/A	Classical-pool comparator
rPPG-10	Reference baseline	26 subjects; source map pending	10.164	16.054	65.4%	100.0%	10.164	34.6%	Canonical subject unit; descriptive comparator
SCAMPS-full [B- SCAMPS-FULL- SUMMARY-v1]	Max-SNR baseline	2,800 synthetic samples	30.234	43.827	N/A	N/A	N/A	N/A	Synthetic boundary comparator; release gate not evaluated in this source row
MCD-rPPG stress	VitalsSight	6 subjects (60 windows); subject-equal endpoint	17.231	24.429	63.3%	45.0%	7.928	7.4%	Supportive stress evidence
UBFC-Phys S1-S14	VitalsSight	45 windows / 5 subjects	12.722	20.170	64.4%	42.2%	3.289	5.3%	Boundary result; conflicting older endpoint retained
UBFC-rPPG classical stress	VitalsSight	439 windows / 42 subjects	8.774	14.710	80.0%	74.9%	5.971	8.5%	Stress evidence; subject- equal MAE 8.746
rPPG-10	VitalsSight	26 subjects; source map pending	7.136	12.285	76.9%	76.9%	2.102	0.0%	Canonical subject unit; supportive stress evidence
SCAMPS-full [B- SCAMPS-FULL- SUMMARY-v1]	VitalsSight	2,800 synthetic samples	29.009	42.939	N/A	100.0%	29.015	54.6%	Negative synthetic boundary

All locked metrics are retained; endpoint-specific units and denominators are stated in the preceding text. N/A in no-gate rows means that release endpoints were not evaluated, whereas the rPPG-10 reference row uses an always-release convention. SCAMPS keys B-SCAMPS-FULL-SUMMARY-v1 and B-SCAMPS-FULL-PAIRED-GATE-v1 are preserved without merging. Shading marks VitalsSight rows; bold is limited to comparable endpoints; heterogeneous protocols are not pooled.

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261 Failure Boundaries and Cross-Modal Ambiguity

262 Mapping the explicit failure boundaries of deep rPPG architectures yielded decisive limitations. Under the synthetic
263 distribution shifts of SCAMPS-full, VitalsSight logged a negative boundary, retaining a 29.015 BPM released MAE with a
264 severe 54.6% error exceedance rate (utilizing the primary B-SCAMPS-FULL-SUMMARY-v1 key). A separate paired-gate
265 evaluation (B-SCAMPS-FULL-PAIRED-GATE-v1) confirmed a 57.3% all-output exceedance with zero max-SNR coverage,
266 reinforcing that synthetic limits currently block safe deployment ([Supplementary Table S13b](#)). Similarly, the rPPG-OSF
267 camera-shift protocol yielded an 84.2% released error exceedance, while the MR-NIRP low-light boundary produced an
268 18.903 BPM MAE with 75.0% exceedance. Dynamic-lighting candidate audits (DLCN) revealed that while a theoretical top-
269 k oracle could achieve a 2.97 BPM MAE, the strongest-power route degraded to a 24.90 BPM MAE (68.54% exceedance),
270 proving that mere candidate availability within the pool is insufficient without a robust selection mechanism.

271 To definitively isolate the mechanics of system failure under extreme low illumination, we conducted an independent audit
272 of 15 released MR-NIRP Indoor RGB/NIR sequences (derived from eight unique participants) using the localized MRNIRP-
273 INDOOR-CASE-AUDIT-v1 protocol with synchronized PulseOx references³³⁻³⁴. Strikingly, 100% of the sequences were
274 routed to review. This withholding was not triggered by an absence of accurate physiological data—every single candidate
275 pool successfully generated a hypothesis within 10 BPM of the clinical reference, with nearest-candidate absolute errors
276 ranging from a highly accurate 0.003 BPM to 1.495 BPM. Instead, all sequences entered the review state because the system's
277 artifact and quality rules detected an unresolvable lack of clean RGB-NIR cross-modal consensus. This explicitly validates
278 the utility of the output contract: rather than blindly outputting an uncalibrated, artifact-driven estimate, the framework
279 correctly identified the signal ambiguity and withheld the autonomous release, ensuring safer operational limits.

280 As visualized in [Figure 6](#), this failure mechanism is highly localized. Cases A1–A2 show stationary apparent agreement
281 across certain bands that nonetheless fails strict quality control parameters. Cases B1–B2 reveal explicit cross-modal spectrum
282 divergence between simultaneous RGB and NIR streams. Cases C1–C2 show that under subject motion, near-reference
283 candidates are successfully preserved within the pool but lack the relational strength to trigger a release. Finally, cases D1–
284 D2 demonstrate motion-driven, low-frequency top-peak locking. Unlike traditional single-route baselines that force an
285 erroneous prediction, VitalsSight successfully surfaces cross-candidate ambiguity and withholds autonomous release when
286 cross-modal consensus degrades. The aggregate findings in [Figure 6](#) support an auditable review contract, though they do not
287 demonstrate autonomous low-light accuracy, beneficial risk control, or unconstrained external generalization. All 15
288 operational decisions and the corresponding eight-case indexing are preserved in [Supplementary Table S13c](#).

289 Beyond low-light boundaries, evaluation of the UBFC-Phys benchmark remains unresolved: baseline all-output MAEs
290 present an unresolved divergence (19.213 BPM vs. 19.665 BPM), while its released summaries report a 2.840 BPM MAE
291 (3.3% exceedance) versus a 3.289 BPM MAE (5.3% exceedance). These conflicting variants and their accompanying window
292 diagnostics are preserved in [Supplementary Table S13a](#). Concurrently, a subgroup accessibility audit executed on the CMU
293 proxy dataset retained 88 records, producing a 77.3% coverage rate, a 5.106 BPM MAE, a 7.4% overall exceedance, and a
294 worst-group exceedance of 9.4%, resulting in a 3.8-point performance gap. It is imperative to note that site or country location
295 contexts function strictly as proxies for data availability; they do not measure formal skin tone distributions and cannot be
296 leveraged to validate algorithmic fairness³⁵⁻³⁷.

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MR-NIRP Indoor: candidate evidence under conservative review

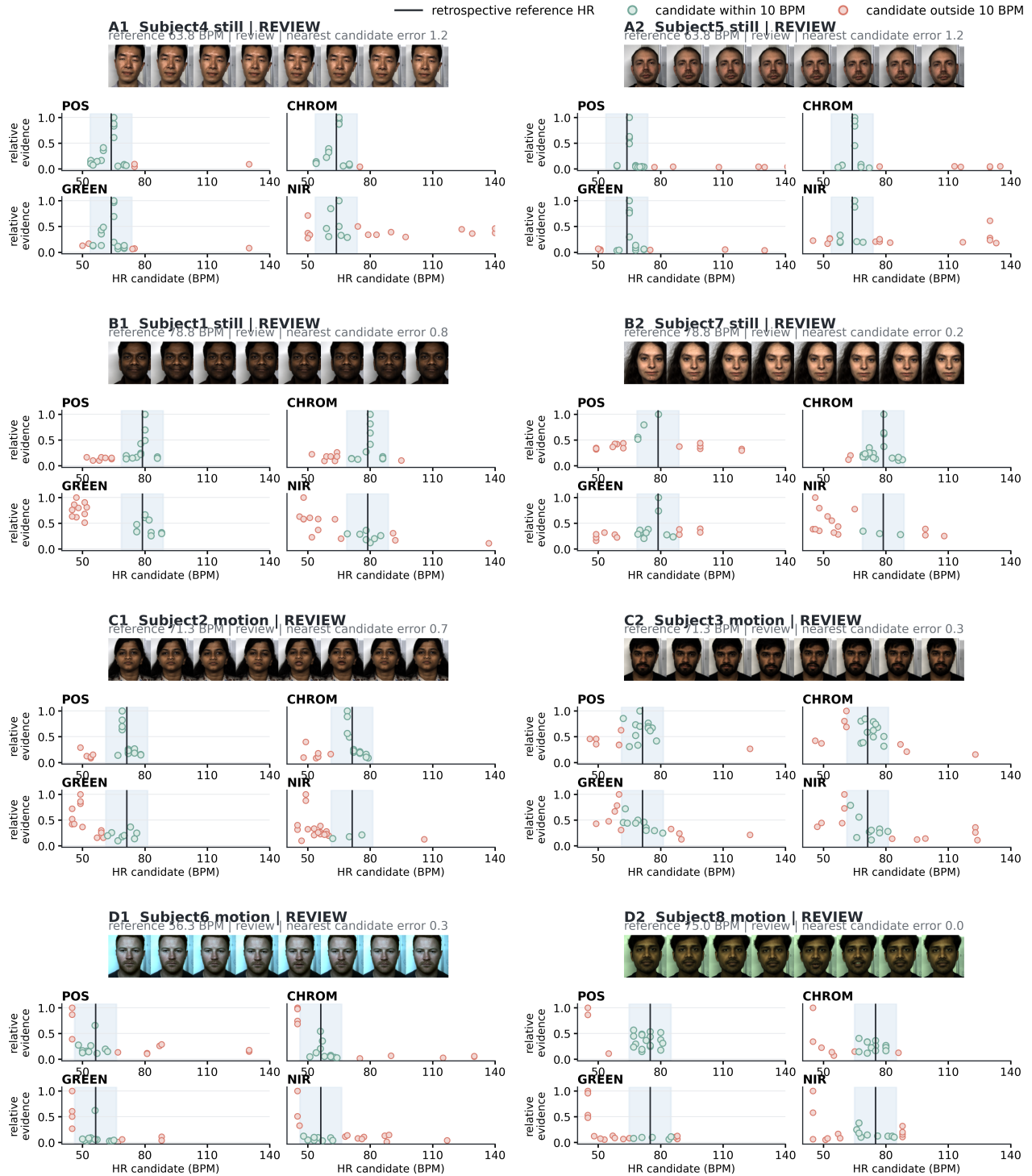


Figure 6 | MR-NIRP Indoor candidate evidence under conservative review. Eight released participants contribute one case each: A, stationary apparent agreement; B, stationary RGB-NIR conflict; C, motion with a near-reference candidate; and D, motion with low-frequency top-peak locking. Panels use the first 60 s and synchronized PulseOx. POS, CHROM, GREEN and NIR peaks are shown against the retrospective reference and an analytical ± 10 BPM band. All 15 audited sequences entered review because clean RGB-NIR consensus was absent, although each pool contained a candidate within 10 BPM. Bayer frames were demosaiced only for display. The source README excluded two participants who declined public release; only the eight released participants

are shown with attribution. This separate protocol does not replace the retained MR-NIRP summary or establish low-light accuracy, beneficial abstention, participant-level risk control or external generalisation.

Computational boundary

Computational profiling conducted on short-window videos resulted in 9 out of 10 attempted cases successfully completing, yielding a total p95 processing latency of approximately 15.7 seconds. As compiled in [Supplementary Table S15](#), stage-specific marginal percentiles measure independent execution boundaries and are explicitly non-additive. Because all downstream deep-backbone predictions were precomputed during our experimental protocols, this latency profile isolates the candidate extraction and relational selection layers; it does not constitute an end-to-end deployment latency or establish real-time operational readiness.

Research web implementation and finite conformance verification

To verify the reproducibility of the system's public software paths, we executed a finite implementation conformance test utilizing seven hash-locked MCD-rPPG video fixtures. These fixtures comprised five original provider recordings, one controlled derivative incorporating artificial face-visibility loss, and one highly truncated 5-second slice. All expected behavioral states were locked prior to execution. Each fixture underwent two direct direct-backend execution replays and one public API serialization run.

Across all 21 executions, VitalsSight demonstrated perfect behavioral conformance, matching the preassigned states to yield exactly 3 releases, 15 reviews, and 3 retake triggers at the execution level. All non-release decisions successfully withheld automated vital sign reporting. The workflow automatically generated 21 linked PDF, JSON, and Markdown evidence packets before strictly deleting raw video uploads from the local analysis root. The single released fixture generated a report value of 75.063 BPM. When matched retrospectively against a reference-only Lead-II ECG value of 77.121 BPM, the estimation yielded an absolute difference of 2.058 BPM, which was never exposed to the underlying inference network.

This clean software revision successfully passed 83 independent mobile/desktop browser assertions across eight parallel digital workspaces, four report exports, and 50 isolated unit tests, generating zero runtime console or unexpected HTTP errors. As detailed in [Supplementary Table S15a](#), this regression matrix formally verifies finite label-free software path routing; it does not substitute for the primary supervised selector end to end, nor does it establish clinical utility, independent cohort accuracy, system security, or deployment effectiveness.

Discussion

The VitalsSight framework successfully reframes a critical failure mode of camera-based heart rate monitoring—silent estimation failure under noise—as a problem of candidate ambiguity, preserving multiple competing pulse hypotheses until a supervised relation selector can systematically compare them²³. Our experimental evaluations validate this core concept: the collective candidate pool achieved lower local window-level errors than any single constituent single-route source, and the selected computational paths varied dynamically across video windows^{12–13,31}. While post-hoc trajectory correction techniques focus on smoothing or constraining a single pre-selected track, VitalsSight operates further upstream; selection determines the foundational physiological branch from the pool. The integration of relation-aware features and explicit harmonic penalties directly addresses structural vulnerabilities like frequency aliases and source-specific camera artifacts. However, because the constituent single-route methods possessed asymmetric training histories, these experimental results cannot be isolated from ordinary ensemble or gating effects. Definitively decoupling selection mechanics from standard assembling requires future benchmarks incorporating perfectly matched candidate pools, standardized computational limits, and identical data split controls.

Our findings strongly indicate that the framework's release/review state is better supported as an auditable output schema than as a mathematically calibrated clinical risk score^{28–29}. Although classical reject-option classification and selective classification methods effectively separate localized model predictions from downstream clinical actions, our empirical risk-coverage curves revealed an adverse diagnostic trend: released subset errors unexpectedly worsened as coverage floors fell. Furthermore, our post-hoc window-level upper confidence bound of 0.084 cannot serve as a statistical guarantee, as it treated highly correlated video windows as independent observations and reused the participant cohort across multiple model configurations. Establishing a defensible clinical risk claim requires a prospectively locked model and decision threshold evaluated via a participant-level estimand on completely untouched participant clusters, using both always-release and simple-rejection mechanisms as standard experimental controls.

The cross-dataset stress testing systematically mapped out the explicit transportability and generalization boundaries of deep remote photoplethysmography (rPPG) architectures. Datasets representing extreme environment or domain shifts—such as SCAMPS-full, rPPG-OSF, and MR-NIRP—retained large, unresolvable error tails under stress. Concurrently, the candidate-pool audit on dynamic-lighting inputs (DLCN) exposed a substantial oracle gap, while the true baseline for UBFC-Phys remains empirically unresolved^{5–6,11,38}. These limitations reinforce how severely ambient low light, elevated heart rates, localized facial-region choices, and overall signal quality alter the availability of physiological evidence. Importantly, site or country location labels must not be leveraged as demographic proxies, as they cannot validate skin-tone fairness or establish true demographic robustness.

While our research web conformance matrix successfully verified automated state routing, selective withholding, reporting, API parity, data persistence, and upload deletion, its underlying engine used a label-free classical bridge rather than the fully supervised selector and deep pool evaluated in our primary experiments. Thus, the web implementation demonstrates that the auditable contract can be functionally instantiated; it does not establish clinical accuracy, safety, user usability, system security, calibrated risk, clinical treatment effects, real-time readiness, or workflow deployment effectiveness^{4,9–10}.

Extending beyond these boundaries to achieve stronger clinical validation requires four strict methodological constraints: (1) matched learned baselines trained under identical participant splits, visual preprocessing pipelines, labels, tuning budgets, and random seeds; (2) participant-equal endpoints accompanied by paired cluster uncertainty intervals rather than isolated

369 window-level metrics; (3) a single, structurally locked prediction source shared across all manuscript figures and tables ; and
370 (4) a prespecified abstention rule evaluated on completely untouched participants and devices.

371 In summary, VitalsSight provides a highly practical framework that couples a candidate-aware estimator with a supervised
372 relation selector and an explicit, auditable review interface while meticulously documenting failures. The primary internal
373 performance of 1.646 ± 0.051 BPM motivates matched evaluation but remains an optimistic window-level estimate lacking
374 participant-cluster uncertainty metrics. Concurrently, the adverse low-coverage behavior prevents any claims of automated
375 release risk minimization. Ultimately, by defining these transportability boundaries transparently, this study narrows the
376 present empirical gaps and renders the validation rules required for safe digital vital sign monitoring fully testable.

377

378 **Methods**

379 **Study design**

380 This retrospective computational study tested candidate retention and whether a post-selection score identified review
381 windows. UBFC-rPPG was primary; [Table 1](#) assigns other datasets retrospective roles. Windows remained nested within
382 participants reported as disjoint across folds per seed. Analyses were not preregistered, no clinical outcome was evaluated,
383 and the participant-first split manifest remains outstanding.

384 **Datasets and participant accounting**

385 UBFC-rPPG contributed 42 participants and 439 analysed windows³⁹. Provider documentation and analysis records support
386 these counts. [Table 1](#) reports units and roles; split hashes and recording-to-window exclusions remain planned. MCD-rPPG
387 contains resting/post-exercise recordings from 600 participants and three cameras under CC BY 4.0³⁰. Stress summaries use
388 60 windows from six participants. The case-prediction crosswalk and candidate-rule-development history remain under
389 reconciliation; values are protocol-specific, not pooled external validation.

390 UBFC-Phys S1-S14 contributed 45 windows from five participants⁴⁰; rPPG-10 retained 26 subjects, 4TU-rPPG 21 sessions
391 and SCAMPS-500/full synthetic samples⁴¹. rPPG-OSF, MR-NIRP, CMU proxies and DLCN defined source-shift, low-light,
392 subgroup or candidate-availability boundaries. The separate 'MRNIRP-INDOOR-CASE-AUDIT-v1' used all 15 released
393 RGB/NIR sequences from eight participants with synchronized PulseOx and does not replace the retained summary³³⁻³⁴.

394 Participant-level splitting crossed model seeds 704, 1704 and 2704 with split seeds 2024-2026. Each split assigned 14
395 participants to training, calibration and testing; calibration sets contained 143-150 windows and tests 140-149. Leakage
396 verification still requires a manifest linking splitting, caching, normalisation and candidate generation.

397 **Ethics and data governance**

398 No new participants were recruited, and no new approval or exemption is asserted. Provider ethics, consent, secondary-use
399 and licence records are not consolidated for every human dataset. The MCD-rPPG report documents participation and data-
400 processing consent, and its dataset record applies CC BY 4.0³⁰. This copyright licence does not establish third-party consent
401 to publish identifiable faces. Nature Portfolio does not treat black eye bars as anonymisation; submission-facing MCD panels
402 therefore remove complete frame strips, while masked versions remain internal pending written confirmation. The MR-NIRP
403 Indoor README states that two participants declined public release and were excluded, leaving eight released participants³³⁻
404 ³⁴. Figures 6 and S3-S4 use only those participants. This provider-reported basis is not new approval or consent obtained by
405 us.

406 **Video preprocessing and window formation**

407 Recordings were decoded and sampled before MediaPipe landmark detection, facial-region formation and RGB/Lab
408 extraction. Reference heart rate was aligned per window only for labels and evaluation. Detection, tracking, filtering, frame-
409 rate, window, stride, padding and alignment remain manifest fields; metrics are not pooled across protocols.

410 **Candidate pool construction**

411 For each window w , the framework retained the variable-size candidate set defined in equation (1), instead of immediately
412 publishing the largest spectral peak.

$$\mathcal{C}_w = \{c_{w,i} = (h_{w,i}, q_{w,i}, a_{w,i}, r_{w,i}, m_{w,i})\}_{i=1}^{K_w} \quad (1)$$

413 Here, K_w is the candidate count and $c_{w,i} = (h_{w,i}, q_{w,i}, a_{w,i}, r_{w,i}, m_{w,i})$ contains heart rate h , source-quality evidence q ,
 414 harmonic/alias evidence a , route identity r and validity/missingness mask m . Their construction must be fixed by the
 415 recovered protocol configuration.

416 Sources included dense facial regions, full/sub-window spectra, harmonic or alias alternatives and correction candidates.
 417 Blind-source, chrominance and spatial-redundancy routes supplied classical evidence ^{12–14,31}.

418 DeepPhys, TSCAN and EfficientPhys supplied learned candidates ^{16–18}.

419 PhysFormer and RhythmFormer supplied transformer candidates ^{19–20}.

420 Framework tooling supplied other implementations ^{22,42}.

421 Each candidate contained a heart rate and categorical source fields. Relations encoded candidate agreement, SNR-like
 422 evidence and harmonic penalties. The audited selector inputs excluded reference heart rate, ground truth and candidate
 423 absolute error. Spectral limits, peak rules, de-duplication, top-k capacity and tie handling remain configuration fields. The
 424 selector cannot recover a physiological branch absent from the pool.

425 Deep candidate generation

426 DeepPhys, TSCAN and EfficientPhys entered as recorded same-window candidate outputs ^{16–18}. PhysFormer and
 427 RhythmFormer entered under the same constraint ^{19–20}. These were not matched retraining benchmarks; commits, checkpoint
 428 hashes, training histories and out-of-fold predictions remain reproducibility fields.

429 Supervised candidate selector

430 Reference-derived labels supervised a variable-set ranker that returned the highest-scoring candidate. The primary branch
 431 used set attention with correction candidates. Reference heart rate and candidate error formed training targets and retrospective
 432 metrics but were absent from audited runtime features; the method is supervised, not reference-free.

433 Equation (2) defines selection from inference-time candidate evidence.

$$s_{w,i} = f_{\theta}(c_{w,i}, \mathcal{C}_w), \hat{i}_w = \underset{i \in \{1, \dots, K_w\}}{\operatorname{argmax}} s_{w,i}, \hat{h}_w = h_{w, \hat{i}_w} \quad (2)$$

434 The set scorer f_{θ} maps candidate and relation evidence to $s_{w,i}$; $\hat{i}_w$ and $\hat{h}_w$ are the highest-scoring index and heart rate.
 435 Tensor shapes, masks and tie handling remain configuration fields.

436 The ranker used seeds 704, 1704 and 2704. Configuration, masking, optimisation and calibration remain reproducibility
 437 fields; evidence supports the selection function, not architecture-level novelty.

438 Correction candidates

439 Optional candidates smoothed predicted sequences without using reference heart rate at inference. Effects differed across
 440 branches, so correction remains a secondary source rather than a validated causal component ²³.

441 Release review and evidence output

442 After selection, equation (3) returned release or review; an earlier quality state could request retake.

$$\rho_w = g_\phi(\hat{h}_w, C_w, \Delta_w) \quad (3)$$

$$d_w = \text{release if } \rho_w \leq \tau;$$

$$d_w = \text{review if } \rho_w > \tau;$$

443 The gate g_ϕ maps the selected estimate, candidate set and temporal evidence Δ_w to ρ_w ; τ separates release from review.
 444 This output notation does not imply calibrated risk control while the gate and confirmatory protocol remain unresolved.
 445 Algorithm 1 states the full sequence. Its evidence packet contains the selected value and source, competitors, agreement,
 446 review reason and model version. This reject-option-inspired schema^{28–29} does not label release as clinically safe.

447 **Algorithm 1 | Candidate-aware heart-rate selection and abstention**

Input: video window V_w ; route set R ; selector f_θ ; gate g_ϕ ; threshold τ

Output: selected HR $\hat{h}_w$; decision d_w ; evidence packet E_w

Initialize: candidate pool $C_w \leftarrow \emptyset$; score list $S_w \leftarrow \emptyset$; evidence $E_w \leftarrow \emptyset$

for each route $r \in R$ **do**

Extract the route-specific ROI trace and spectral evidence from V_w

Generate the primary HR candidate and plausible harmonic or alias alternatives

Append each candidate tuple to C_w

end for

if $C_w = \emptyset$ **then**

Return review with an insufficient-candidate reason

end if

for each candidate $c_{w,i} \in C_w$ **do**

Compute agreement, route evidence, temporal plausibility and harmonic risk

Compute $s_{w,i} = f_\theta(c_{w,i}, C_w)$ and append the auditable fields to E_w

end for

Select: $\hat{i}_w = \arg \max \text{ over } i \text{ of } s_{w,i}$ and $\hat{h}_w = h_{w,\hat{i}_w}$

Compute: $\rho_w = g_\phi(\hat{h}_w, C_w, \Delta_w)$

if $\rho_w \leq \tau$ **then** $d_w \leftarrow \text{release}$ **else** $d_w \leftarrow \text{review}$

Update: E_w with selected route, agreement, gate score, reason and model version

Return: $\hat{h}_w$, d_w and E_w

448 A review decision withholds autonomous release while retaining evidence for audit. Reviewer roles, storage, access control,
 449 retention, retake limits and escalation were not evaluated.

450 **Training validation and leakage controls**

451 Seeds 704, 1704 and 2704 used participant-grouped splits. Primary mean $\pm$ SD values remain unchanged although per-seed
 452 rows are absent. Runtime excluded reference heart rate, ground truth and candidate error; train/test matrices were standardised
 453 separately. A final manifest must link participant, window, checkpoint, prediction, seed and split across all stages.

454 Accuracy release and threshold endpoints

455 Absolute error was the difference magnitude between selected and aligned reference heart rates. Equations (4a) and (4b) define
 456 all-window MAE and RMSE, respectively; equations (5)-(7) define coverage, released MAE and released threshold
 457 exceedance.

$$\text{MAE} = \frac{1}{N} \sum_{w=1}^N |\hat{h}_w - h_w^*| \quad (4a)$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{w=1}^N (\hat{h}_w - h_w^*)^2} \quad (4b)$$

$$\text{Coverage} = \frac{1}{N} \sum_{w=1}^N \mathbf{1}[d_w = \text{release}] \quad (5)$$

$$\text{Released MAE} = \frac{\sum_w \mathbf{1}[d_w = \text{release}] |\hat{h}_w - h_w^*|}{\sum_w \mathbf{1}[d_w = \text{release}]} \quad (6)$$

$$\begin{aligned} &\text{ReleasedExceedance}_\epsilon \\ &= \frac{\sum_w \mathbf{1}[d_w = \text{release}] \mathbf{1}[|\hat{h}_w - h_w^*| > \epsilon]}{\sum_w \mathbf{1}[d_w = \text{release}]} \end{aligned} \quad (7)$$

458 Released metrics with no released window are N/A, not zero. Zero denotes an observed zero proportion; N/A denotes an
 459 undefined or inapplicable endpoint. Not reported denotes a heterogeneous aggregate without one route-specific value. In
 460 equations (5)-(7), $\mathbf{1}[\cdot]$ is the indicator function, h_w^* is the aligned reference heart rate, and epsilon is the analytical error
 461 threshold. The main threshold was 10 BPM, with sensitivity at 5, 8, 10 and 15 BPM; none is clinical. Subject-equal endpoints
 462 averaged within-participant values. The retained 1.662 BPM lacks per-seed components but remains unchanged. No-release
 463 participants contribute zero coverage and undefined conditional error; the headline remains window level.

464 Exploratory risk audit

465 The entropy-margin audit used margin, entropy, candidate count, harmonic risk, agreement and source category across three
 466 seeds and three 14/14/14-participant splits. At $\alpha = 0.09$ and $\delta = 0.05$, a one-sided Clopper-Pearson bound treated
 467 windows as independent. Clustering and post-hoc choices make 0.084 exploratory.

468 Statistical analysis

469 Seed summaries are mean $\pm$ standard deviation across three model seeds, not participant confidence intervals. No primary P
 470 value, paired cluster contrast or superiority claim is reported. Confirmation requires participant-level paired resampling with
 471 prespecified replicates, interval, sidedness and multiplicity.

472 Ablation and boundary analyses

473 Ablations compared relations, harmonic terms, soft targets, correction candidates and release rules in separate graph-selector
 474 or correction branches. They retain branch labels and no paired participant significance claim. Stress datasets retain local
 475 protocols and are not pooled. Submission-facing MCD facial panels were excluded because image-publication permission
 476 was not established; consent-filtered MR-NIRP panels remain descriptive.

477 **Runtime and research web verification**

478 The runtime audit retained nine completed cases and one failed UBFC-rPPG worker (return code -6). It measured decoding
479 through serialisation. Total p95 was 15,666.001 ms; marginal percentiles are not additive. Precomputed deep candidates
480 excluded backbone inference, so this is not end-to-end latency. Hardware, warm-up, memory and software versions remain
481 reproducibility fields.

482 Research web verification used seven hash-locked fixtures with preassigned states: five provider videos, one controlled
483 visibility derivative and one 5-s acquisition-gate slice. Two backend and one API run per fixture produced 21 records checking
484 state parity, withholding, report consistency, upload deletion and API behavior. Browser tests covered release, review and
485 retake at 1440 x 1000 and 390 x 844 pixels, 83 assertions, eight workspaces and four report formats. Services were bound to
486 the clean commit/tree and designated empty-after-analysis upload root; 50 unit tests passed. Lead-II ECG was a reference-
487 only comparison for the released fixture. This is a curated conformance replay, not an independent accuracy cohort, full
488 supervised/deep end-to-end test or clinical-workflow validation.

489

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491 Not applicable.

492

493 **Author contributions**

494 Y.Wu., Z.C., K.L., Z.B., R.M.E., and Y.Wang. conceptualized and designed the study. All authors had full access to the data
495 and contributed to the decision to submit the manuscript for publication. Y.Wu., Z.C., K.L., Z.B., R.M.E., and Y.Wang.
496 contributed to manuscript drafting. Y.Wu., Z.C., K.L., performed the data analyses. All authors reviewed the manuscript
497 critically for important intellectual content and approved the final version. Data collection was performed by Y.Wu., Z.C.,
498 K.L.

499

500 **Availability of data and materials**

501 Research code for candidate generation, ROI aggregation, candidate selection, release/review, evidence-linked reporting,
502 participant-cluster analysis, risk analysis and partial runtime profiling is publicly available at
503 <https://github.com/ShikiRyo1/VitalsSight-candidate-aware-rPPG>. The release includes dependencies, selected configurations,
504 a dataset-free example, protocol descriptors, aggregate metrics and the browser/API validation harness. Raw videos,
505 identifiable frames, third-party datasets, provider-controlled references and checkpoints are excluded. This research artifact is
506 not a validated clinical system; no software license, release tag or DOI has been assigned.

507

508 **Declarations**

509 The authors declare that they have no conflicts of interest regarding the publication of this work.

510

511 **Software and reproducibility**

512 Records retain model seeds 704, 1704 and 2704, split seeds 2024-2026, build scripts, figure sources and tables. The public
513 release adds executable analysis code and dependencies; provider-controlled data, frozen subject splits and checkpoints remain
514 necessary for end-to-end reproduction.

515 **Data availability**

516 Third-party datasets may be redistributed only where provider licenses permit. Aggregate metrics and protocol descriptors
517 accompany the public code release. Raw videos, identifiable frames and provider-controlled references are excluded. No
518 permanent data DOI is available, and some license records remain incomplete.

519

520 **Consent for publication**

521 Not applicable.

522

523 **Competing interests**

524 The authors declare that they have no conflicts of interests.

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